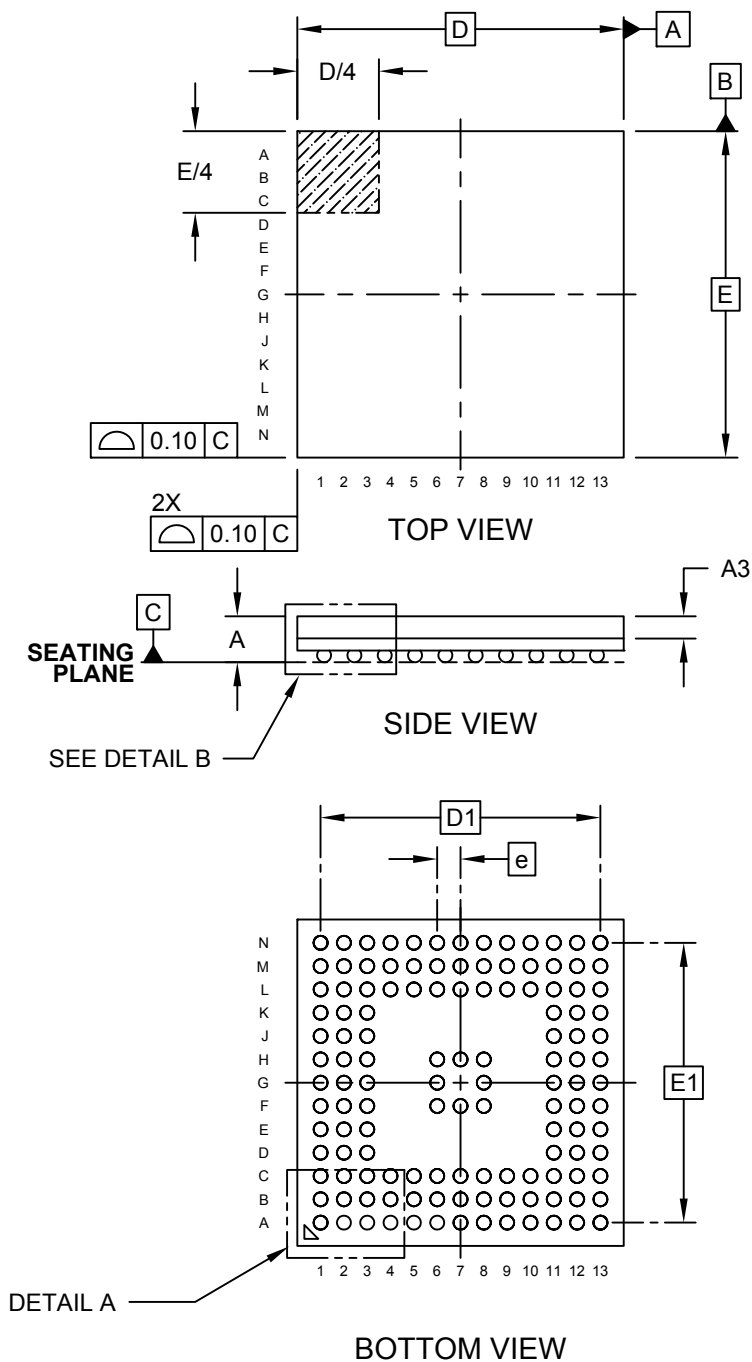


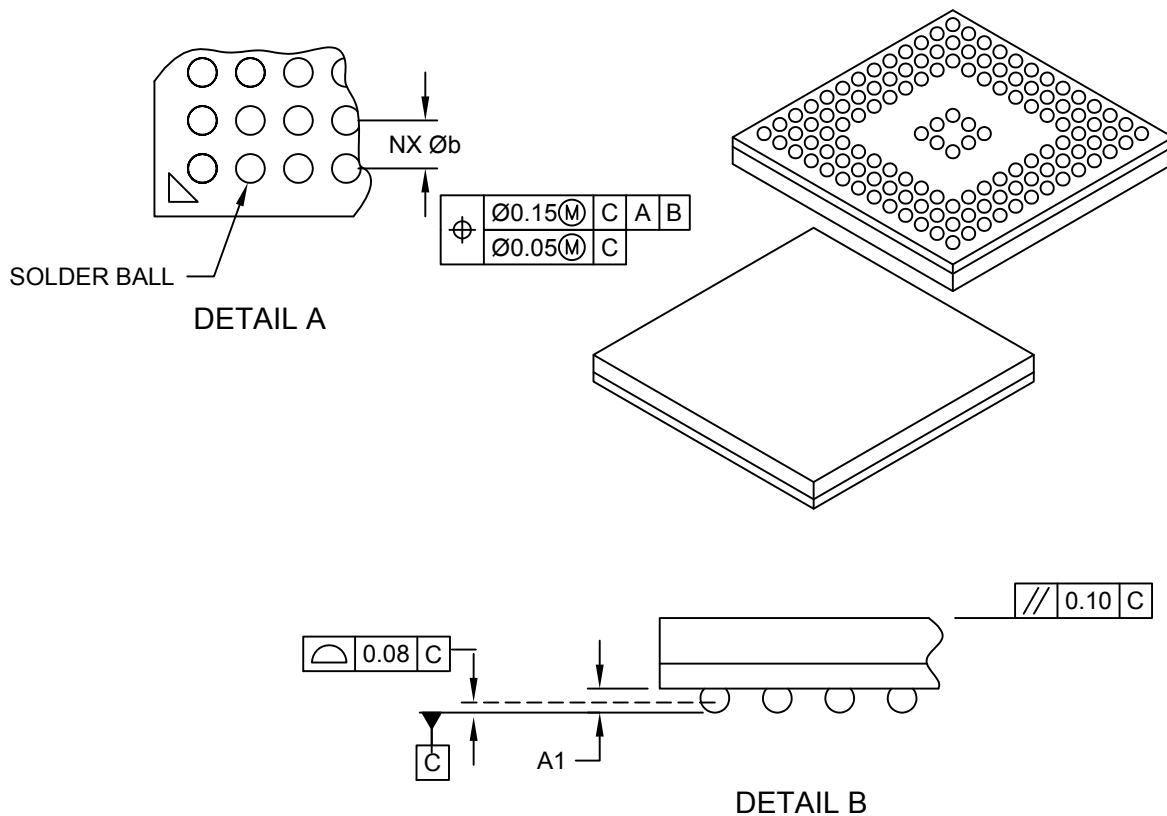
128-Ball Thin Fine Pitch Ball Grid Array (4H) - 7x7x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



128-Ball Thin Fine Pitch Ball Grid Array (4H) - 7x7x1.2 mm Body [TFBGA]

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Pins	N	128		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.16	-	0.26
Molded Cap Thickness	A3	0.530 REF		
Overall Width	E	7.00 BSC		
Overall Ball Pitch	E1	6.00 BSC		
Overall Length	D	7.00 BSC		
Overall Ball Pitch	D1	6.00 BSC		
Ball Diameter	Øb	0.25	0.30	0.35

Notes:

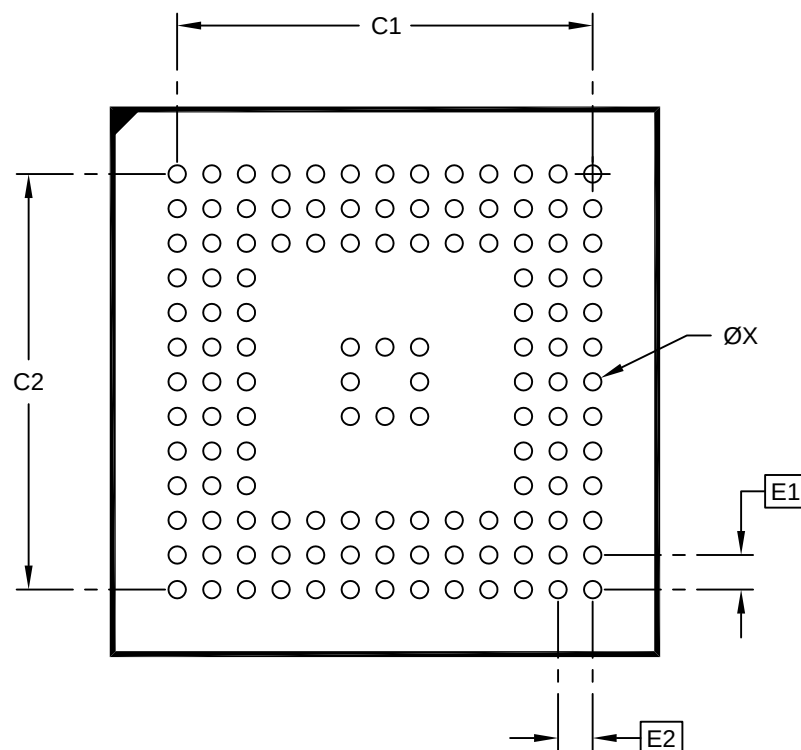
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

128-Ball Thin Fine Pitch Ball Grid Array (4H) - 7x7x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E1	0.50 BSC		
Contact Pitch	E2	0.50 BSC		
Contact Pad Spacing	C1		6.00	
Contact Pad Spacing	C2		6.00	
Contact Pad Diameter (X128)	X		0.25	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.